Sahaj Bhadja

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Summary

Software Engineer with 2+ years running production services and **databases** on **AWS** and Azure, automating **Linux** build and test pipelines, troubleshooting performance, and monitoring with Sentry and Kibana, scripting in **Python**, Java, and JavaScript.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Troubleshot **database performance** on a multi-tenant Azure SQL and PostgreSQL platform, redesigning indexes and access patterns to cut dashboard latency from 1.2s to 350ms for 500+ users.
- Operated recurring production jobs with failure isolation, explicit job states, and reconciliation checks, sustaining 99%+ **reliability** across 5K+ monthly integrations and 1K+ AI jobs with no cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Supported a production backend on **AWS** with MySQL on RDS serving 150+ instructors and 10K+ daily requests, keeping API response stable during peak load through concurrent batch processing.
- Added timeout-aware failover across external model providers and per-call usage and cost logging for **monitoring**, shipping releases through GitHub Actions CI/CD to Render and Vercel production deployments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated nightly **Jenkins** jobs that allocated VMs, pulled commits, built and installed executables, and ran tests across Linux, Windows, and macOS, emailing failure logs and coverage reports.
- Coordinated parallel builds across India and US servers with a **global team**, and troubleshot large-scale API integrations with indexing and caching, improving response times 30% and cutting storage costs 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Ran production **observability** with Sentry for errors, Azure Monitor for performance metrics, and Kibana for logs across eight Node.js and MongoDB microservices synchronizing 2M+ records across multiple locations in real time.
- Provisioned infrastructure with ARM templates on **Azure** App Services and shipped Docker builds through Jenkins CI/CD with approval gates and rolling zero-downtime updates at 95%+ build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated Python Flask APIs with existing **AWS**-hosted authentication and geo-data microservices and Firebase, sustaining sub-200ms latency across 50K+ booking records and using transactional locking to prevent double bookings.
- Automated payment event handling with idempotent **webhooks** for refunds, disputes, and chargebacks, cutting manual work 70%, and scheduled hourly Databricks Spark jobs over 50K+ booking and payment events.

Skills

Languages & Scripting: Python, Java, JavaScript, TypeScript, Bash, Shell, SQL
Systems & Networking: Linux, Windows, TCP/IP, NGINX, REST APIs, WebSockets, Performance Troubleshooting
Databases: MongoDB, PostgreSQL, SQL Server, Azure SQL, MySQL, Indexing, Query Optimization
Cloud & Infrastructure: AWS (EC2, S3, RDS, ECS/Fargate), Microsoft Azure, GCP, Docker, Kubernetes, ARM Templates
Monitoring & Automation: Sentry, Azure Monitor, Kibana, Grafana, Jenkins, GitHub Actions, CI/CD, Alerting
Practices: Incident Troubleshooting, Root Cause Analysis, Git, Firebase, Databricks, Agile/Scrum

Projects

Database Incident Triage Toolkit | *Python, MongoDB, Prometheus, Grafana, Bash, AWS*

- Designed a Python toolkit that polls MongoDB replica-set status, slow-query profiler output, and host metrics, raises **Grafana** alerts on lag or connection spikes, and runs scripted first-response **runbooks**.

FinNexus Live | *MongoDB, Kubernetes, Next.js, TypeScript, WebSockets*

- Managed Kubernetes deployments and optimized **MongoDB** data access under high transaction volume for a fintech platform serving 50K+ users and 1M+ transactions with CI/CD automation and real-time updates.

Linguistix | *Python, Docker, AWS ECS/Fargate, CodePipeline, CI/CD*

- Containerized and deployed a Python REST data processing API on **AWS** ECS/Fargate with automated CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade operational reliability.